

eyemR-Talk: Using Speech to Visualise Shared MR Gaze Cues

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ABSTRACT

In this poster we present *eyemR-Talk*, a Mixed Reality (MR) collaboration system that uses speech input to trigger shared gaze visualisations between remote users. The system uses 360° panoramic video to support collaboration between a local user in the real world in an Augmented Reality (AR) view and a remote collaborator in Virtual Reality (VR). Using specific speech phrases to turn on virtual gaze visualisations, the system enables contextual speech-gaze interaction between collaborators. The overall benefit is to achieve more natural gaze awareness, leading to better communication and more effective collaboration.

KEYWORDS

Mixed Reality remote collaboration, gaze visualization, speech input

ACM Reference Format:

Allison Jing, Brandon Matthews, Kieran May, Thomas Clarke, Gun Lee, and Mark Billingham. 2021. *eyemR-Talk: Using Speech to Visualise Shared MR Gaze Cues*. In *SIGGRAPH Asia 2021 Posters (SA '21)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3476124.3488618>

1 INTRODUCTION

During face-to-face task collaboration, people can use non-verbal cues such as gaze or hand gestures to support verbal communication to build common ground. However, when collaborating remotely on a physical task using audio or video calls, non-verbal cues may not be conveyed as naturally as in a face-to-face setting.

Researchers have used live 360° panoramic Mixed Reality (MR) to enable remote users to feel immersed in the same task space [Lee et al. 2018]. Recent studies have also investigated sharing virtual gaze cues in MR collaboration and have found that this helps align the collaborator's current focus and enhances mutual understanding [Jing et al. 2021b]. Using virtual gaze cues to share gaze visualisations also encourages the active use of deictic references (e.g. "this" or "that") to facilitate coordination in the remote task space. Deictic references are spoken phrases such as "this one" while

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SA '21, December 14–17, 2021, Tokyo, Japan

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ACM ISBN 978-1-4503-8687-6/21/12.

<https://doi.org/10.1145/3476124.3488618>

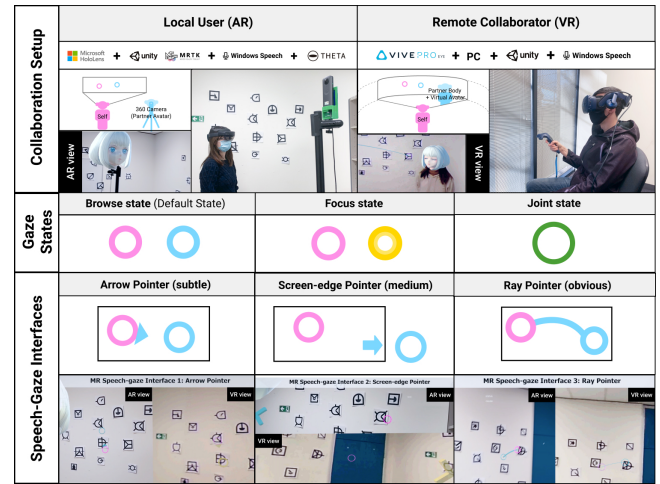


Figure 1: *eyemR-Talk* system overview: Illustration and demonstration of the system setup, gaze states, and shared gaze indicator interface designs

the user is looking at a location of interest. Researchers have found that deixis is successfully acknowledged more often and faster with gaze visualisations [D'Angelo and Begel 2017]. However there has been little research on how verbal cues can be combined with gaze visualisation in MR remote collaboration.

In this poster we present *eyemR-Talk*, an MR remote collaboration system that integrates both verbal and gaze cues to naturally represent shared interests in collaborators' conversations. Our work combines and extends the previous research in gaze visualisations and MR remote collaboration, aiming to provide additional interaction techniques to encourage proactive task coordination and mutual understanding. The main novelty is that we analyse speech input to automatically trigger gaze visualisations to support effective collaboration. The goal of this research is to provide a natural way to connect users' current focuses based on their conversation, resulting in explicit visual guidance to encourage joint attention, therefore improving mutual understanding.

2 THE EYEMR-TALK SYSTEM

Evolved from our previous system [Jing et al. 2021a], *eyemR-Talk* is a speech controlled bi-directional shared gaze system that integrates 360° panoramic live video into MR remote collaboration.

2.1 360° Mixed Reality Remote Collaboration

The local user in the physical task space wears an AR head-mounted display (HMD) enabling them to interact in the immediate surroundings. The remote collaborator wears a VR HMD to view the shared scene and exchange interactions with the local user. Both AR and VR HMDs have built-in eye-trackers to capture gaze data and we use Windows Speech Recognition¹ to detect voice input. The user views are independent shared via a live 360° panoramic camera.

We use the HoloLens2² see-through AR HMD and its built-in eye-tracker to show visualisations for the local user. The Ricoh Theta V³ 360° panorama live camera (1080p) is used to capture the task space and a Vuforia⁴ tracking marker is attached to the camera to align the AR-VR coordinate systems. The Vive Pro Eye VR HMD⁵ with built-in eye-tracker is used by the remote collaborator. The transmitted 360 degree scene of the task environment is mapped onto a sphere which the VR user views from the centre. The VR application runs on a PC (Intel Core i7-6700 3.40GHz CPU, 16GB DDR4 RAM, NVIDIA GeForce 950 GPU) using Windows 10 OS. The system is developed on the Unity game engine. All of the visualised information (gaze data, HMD positions, avatars) are exchanged via Wi-Fi to be displayed synchronously between users.

2.2 Speech-Gaze Interaction

Real-time Gaze Representation: We added two virtual avatar heads to simulate the collaborators' head movements. In the local user's AR view, the remote collaborator's virtual avatar head sits where the 360° camera is placed, whereas the remote VR collaborator views a virtual avatar head superimposed on top of the local user's HMD in the live video view. Both collaborators see their gaze points visualised as circular cursors.

Shared Gaze Visualisations. : The user's own gaze indicator stays in a default state to minimise distraction, but their partner's gaze is presented differently based on their gaze behaviours. In Fig 1 there are three gaze behaviours: (1) Browse state-active eye movements such as saccades or shifts in blue; (2) Focus state-if the user's gaze dwells for 500ms or more, a double ring cursor is shown in yellow. (3) Joint state-when the two collaborators' gaze overlaps, the gaze indicators will be replaced with a green cursor twice as large to indicate shared gaze location.

Speech Input. : To build an explicit connection between verbal conversation and non-verbal gaze visualisations, we used the Windows DictationRecognizer API to inspect contextual information during task coordination. We experimentally added six frequently used deictic references and keywords into an input list: "this", "that", "here", "there", "look" and "see". When either user speaks one of the target keywords, the system guides the collaborator who receives (receiver) the verbal communication to attend to the partner who initiated it (sender). This is done by showing shared gaze indicators in the receiver's view when a phrase in the keyword list is spoken by the sender. This visualisation explicitly connects current gaze focuses between collaborators, therefore the combination of

contextual keywords and gaze cues together encourages joint gaze interactions.

Speech-Gaze Interface. : Constantly pointing towards the partner's current gaze focus when shown, the shared gaze indicator is designed to guide collaborators to achieve joint attention when one of them verbally signals the other during the task coordination. The cue disappears either when a joint state is reached or after a time threshold (5 seconds). There are three different MR visualisations developed to showcase shared gaze indicators depending on the visibility spectrum (Fig 1 under "Speech-Gaze Interfaces"): (1) Arrow Pointer (subtle) - An arrow that rotates around the user's own gaze cursor to point at the partner's cursor; (2) Screen-edge Pointer (medium) - when the partner's gaze cursor is out of the field of view (FOV), a bigger arrow appears along the edge of the user's FOV to point towards the partner's cursor; (3) Ray Pointer (obvious) - A curved ray links between the user's own gaze cursor and the partner's gaze cursor to guide the user to the partner.

2.3 Use Cases

By improving the connection between verbal and gaze communication cues in MR remote collaboration, our approach is suitable for physical tasks where the non-verbal cues are less accessible or for users where physical cues are temporarily disabled. For example, hand pointing during surgery or gesturing during machine assembling.

3 CONCLUSION

In this paper, we present eyemR-Talk, a 360° MR remote collaboration system that uses speech input to visualise shared gaze cues in three different styles with three levels of visibility. By visualising gaze cues triggered between conversing remote collaborators, our system connects verbal communication with non-verbal gaze behaviours naturally within the context, providing novel interaction methods to achieve joint attention and mutual understanding. We plan to further explore this concept through formal user studies.

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